Intro 001 — the script, as cut

12:23 · 60 picks · 23.976p · rough cut at `build/roughcut/intro001.roughcut.fcpxml`

This is not the transcript of the tapes. It is a read-back of the **assembled cut** — transcribed off the rendered audio and then punctuated and paragraphed by hand. So it is what the video currently says, in order, not what the source said or what anyone meant to say. Where the two differ, the audio wins; see [S-026].

Only her voice is in the cut. Nathan's side of the conversation, the slates, the caramel in her teeth and the "did you hit record" are all out.

Source	28:43 across four clips
Cut	12:23 (43%)
From MVI_0019	28 picks, 6:06 — the history and the biology
From MVI_0021	28 picks, 6:03 — the practical half
From MVI_0016	4 picks, 0:14 — the scripted opening
From MVI_0018	nothing — see "What was left out"
Music	*Twin Lakes* under the hook only, fading out across 0:22–0:32; *Two Big Reasons* from 11:51. Both 16 dB under her.
Intro sound	The four MVI_0016 shots are lifted **+19.7 dB** — they were recorded that far under the rest

Timecodes are timeline position in the rough cut. ⚠ marks something that needs a decision or a verification before this gets locked; every one of them is listed with a fix in `build-sheet.md`. **Picture** notes say what is on screen and what should be — they are the same items as §B of the build sheet, said where the words are, so a shot can be argued for against the line it has to serve.

Cold open — 0:00

> This is yeast bubbling in action.
>
> This is me, way too excited about yeast.
>
> I'm Katie, I'm a food scientist, and this is my deep dive into yeast.

⚠ §9 wants the hook to make a concrete claim about something home cooks get wrong, inside 15 seconds. This promises a subject but doesn't name a mistake. It also runs 13.8 s before the first idea lands.

Picture. *Shot 1 is now the starter jar rising* — the 4K clip, sped 3.37× to fill the line and deflickered twice so the auto-exposure hunting stops reading as a strobe. *Shot 2 is Katie waving*, lifted from between takes at 0:34 in the same roll. Shots 3 and 4 are still her to camera.

Still wanted: the jar shot wants a **macro** partner — bubbles breaking at the surface, side-lit, shallow enough that the back of the jar goes soft. And **nothing in the graphic language has appeared yet**: §9 asks for the first graphic inside 30 seconds, which means it belongs under "deep dive into yeast" — the title card is the natural candidate.

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1. Nobody knows when it started — 0:13

> As humans, we started using yeast thousands of years ago.
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> The records for when yeast first showed up is kind of murky, from what I could find in my research. About 3,000 years ago, about 1,000 BC — we see some in Egypt, buried with the pharaohs. They have jars, yeast on the side, from what the DNA test can tell.
>
> We also have yeast showing up in the archaeological records from Asia into Europe, all over the place. We don't have exact ways of knowing which one was first, where they were, if we got all of the records — but we know it's all over the place. It's hard to pull out the DNA, because it's also mixed in with a bunch of other things, from my understanding.

⚠ The date, and "buried with the pharaohs". ⚠ She says "DNA test can solve"; read as "can tell".

Picture. Her to camera throughout, which is the longest unbroken stretch in the video after the enzyme section.

Graphic — the timeline card. 1000 BC on the left, "all over Asia and Europe" as a spread rather than a point. The honest version of this graphic is *vague on purpose*: she says the records are murky, so a crisp date pin contradicts the line it sits under. Torn edge, no end-cap.

B-roll. The pharaoh jars are the one place archive footage would earn its keep — a museum storeroom, a sealed vessel. If that can't be licensed, this is a **drawn card**: a jar in section with the residue at the bottom, which is what the DNA came from.

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2. Passed hand to hand — 0:55

> Then we have this understanding of yeast. It's being passed between brewers and bakers. It's being collected from the air. It's being collected from outside of grapes and introduced into wine. It's moving around — it's passed from person to person, baker to brewer, brewer to baker. It's in mead. It's in wine. It's in beer. It's in our sourdough starters. It's just all around.
>
> And what they're doing is they're passing it from successful batch, passed on to someone else's, and they make a successful batch. And if it's a dud batch, they don't pass it on — because who wants to do anything with a dud batch?
>
> And so what we're getting is selective breeding of good yeast. Yeast that ferments, that makes the bread bubble, makes the alcohol happen, is being passed around. And that is what we've used for years and years and years.

⚠ The transcript hears "It's in meat"; in context this is almost certainly "mead" — the same word she uses on the label in §3. Confirm by ear before captions.

Graphic — the exchange. Two hands, a jar passing between them, and the same jar coming back. Cardboard cutouts, stepped on 2s. The dud batch gets the ✗ from the standing mark library and is pushed out of frame rather than fading.

B-roll. Wild yeast on the outside of grapes — a macro of the bloom on a dark grape is one of the few genuinely beautiful shots available here, and she names it out loud. Also: a sourdough starter being fed, which recurs three more times later and is worth shooting once, properly, from several angles.

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3. Pasteur, and then a name — 1:53

> And it was just thought it was spontaneous generation. That's what it was, until we get to Louis Pasteur. Louis Pasteur comes in and he does his gooseneck bottle experiment, and proves that it's not spontaneous generation that is making food come alive — it is bacteria, it's fungi, it's a bunch of microscopic little things in our food. And that's it. That's all we know, for a while, about yeast.
>
> And then we get Franz Julius Ferdinand Meyen, in 1838. He comes in and he discovers what we know for most of our yeast today: *Saccharomyces cerevisiae*. And that is the yeast that we see in our active dry yeast. It's what we find in our fresh yeast, our instant yeast, and most of our sourdoughs.

⚠ Name, spelling and date, and what exactly is being credited to him — she stops on tape to check the name and still lands on "Mayen". This is the single most checkable claim in the video.

Graphic — Pasteur's gooseneck flask. The single most drawable object in the video: the S-bend is instantly readable and nothing else looks like it. Built or drawn, not photographed.

Graphic — the name card. *Franz Julius Ferdinand Meyen · 1838 · Saccharomyces cerevisiae.* This is the one card that has to be typographically perfect, because it is three proper nouns and a date, and it is on screen while she says them. **Blocked on D2** — the spelling in the audio is wrong.

B-roll. Nothing needed. This section is carried by two graphics and her delivery.

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4. It is all the same organism — 2:39

> I have marketing for yeast for mead. It's *Saccharomyces cerevisiae*. I have instant dry yeast. It's *Saccharomyces cerevisiae*. I have active dry yeast with a little sugar in it — guess what? *Saccharomyces cerevisiae*. Fresh yeast: *Saccharomyces cerevisiae*. I even have what could be the start of a sourdough starter — here, it's a pre-ferment. It is yeast, *Saccharomyces cerevisiae*, mixed with flour and water.
>
> It is like 90% of all the yeast that we use is this one bug.
>
> And in sourdoughs — it's in most of the sourdoughs, but sourdough is where we see it change from batch to batch, mother to mother. There might be some other yeast inside of those batches, and they

> have other bacteria, and that's what makes sourdough just a little bit different than your everyday yeast-flour pre-ferment like I have here.

⚠ The 90% figure. This is the best sequence in the footage and it is carried entirely by what she is holding — it needs the insert shots, see the build sheet.

Picture — this is the insert sequence, and it is the best 24 seconds in the footage. She picks up five containers in a row and names each one. Five clean close-ups, cut on her handling: mead yeast, instant dry, active dry with sugar, fresh yeast, the pre-ferment jar.

Shoot them on the same table in the same light so they cut straight into the wide. Label legibility is the whole point — if the type on the packet doesn't read at 1080p, the joke doesn't land.

Graphic. The species name appears five times in twenty seconds. That is a gift: set it once as a card and bring it back identically each time, so the repetition becomes the visual argument rather than a subtitle. Then **90%** as a single large number over the group shot.

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5. What yeast actually is — 3:27

> So what is yeast, in and of itself? Yeast in and of itself is a small little bug. It's a fungi. It's a single-cell organism. It's eukaryotic.
>
> It reproduces itself by budding, where a mother cell will start to grow a little bud off of it for a daughter cell. And the daughter cell, once it gets big enough — mitosis happens, and it splits, and then it continues to grow, and those split, and those split. And that is how *Saccharomyces cerevisiae* splits and grows and reproduces.
>
> All over — we use it everywhere, from our home bakeries to the brewers down the road to the vineyards in California and France. Everywhere. *Saccharomyces cerevisiae*. Same guy, again and again and again.

Graphic — the budding cycle. Mother cell, bud, daughter, split; then those two split. This is the molecular kit's first real job. Four to six variants of each cell so no two are identical, stepped on 2s, and the split should land on a beat in her delivery rather than on a timer.

B-roll. A microscope shot of live budding yeast is the obvious want and the hardest to get. Worth checking whether a university lab nearby will let you point a camera down an eyepiece for an afternoon — it would carry this entire section and could be reused forever.

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6. What it needs — 4:24

> So it will bud and reproduce in an environment that it thrives in. It likes it to be nutritionally dense for it — that's where it likes having starches and sugars to break down, and give the energy that it needs to continue to grow.
>
> It likes its temperature between 54 degrees Fahrenheit and just over 100 degrees Fahrenheit.

> Underneath 54 degrees, it goes back into a little shell of itself and hibernates. It just tucks away and takes a nap until it warms back up and is a good environment for it to grow and thrive. And that's how it survives not being able to move from environment to environment.
>
> Over 100 degrees Fahrenheit it can still grow, until it gets to about 140 degrees Fahrenheit — and that's when baking happens, and it kills it off in the oven. You might get oven spring from the yeast still growing, and other things happening, but then it stops. And you're not eating fresh yeast when you eat bread, because you've killed it off.

⚠ 54 °F / 100 °F / 140 °F. Three numbers, one graphic, all three need her sign-off.

Graphic — the temperature scale. One vertical strip: 54 °F at the bottom, "just over 100" as the happy band, 140 °F as the kill line. The thermometer cutout is already in the §1.1 kit list. Numbers on screen mean **D3 has to clear first**.

B-roll. Dough going into a hot oven, and the rise happening in the first minutes — oven spring is a thing she names that almost nobody has actually *seen*. A glass-door oven and a timelapse would do it.

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7. What it does — the chemistry — 5:24

> So you have your starch, and it breaks down with alpha amylase and beta amylase — and we'll get into that in another video. And then that breaks down to sucrose. And sucrose exists around the yeast, and the yeast itself provides invertase. And invertase is an enzyme — if it ends in "-ase", it's an enzyme. And so the invertase goes in to sucrose and breaks it down into glucose and fructose.
>
> You get zymase, and it breaks it down into your CO_2 and your alcohol. That's your carbon dioxide and your alcohol.
>
> And then when you bake the bread, the alcohol and the CO_2 kind of burn off and leave the bread, so they're not left behind in the bread after you've cooked it. And you also have your CO_2 — that's where you get your bubbly wines, like your champagne.

⚠ **This section is the weakest thing in the footage, and she says so on tape.** Immediately before the first usable take she says "*I think this is where I'm missing — my notes are not complete enough*", and she qualifies the invertase step with "*I believe*". Three separate things are wrong or missing here and none of them are the pipeline's to fix:

1. **Starch → sucrose is not a step.** Amylases cut starch toward maltose; the sucrose in a dough is mostly what was already in the flour or what you added. As cut, it says the amylases produce the sucrose that invertase then splits.
2. **The payoff line is missing.** She reaches "*and that is why we use yeast in our foods — because those are the byproducts…*" and trails off without finishing it. It is cut, so the section ends on champagne instead of on the point.
3. **The alcohol beat is garbled.** "*But the alcohol is left behind into… your alcohol*" — the contrast she wants (it bakes off in bread, it stays in beer and wine) never lands. Cut.

The rest of the video works without this section. It should be re-shot against confirmed notes rather than patched in the edit.

Picture. Her to camera, reading from notes, and it shows — this is the section she flags on tape.

Do not build these graphics yet. The chain as cut has a factual problem (D4) and a hole where the payoff line should be (C2), and a diagram would harden both. Once it is re-recorded, this becomes the molecular kit's centrepiece: starch, the amylases, sucrose, invertase, glucose and fructose, zymase, and CO_2 and ethanol arriving as the two things you can actually see and taste.

B-roll, whenever it is reshot. Bubbles under a loaf's skin. And champagne — she names it, and it is the one place in the video where the payoff is a glass of something.

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8. What slows it down — 6:20

> In your dough, you have fats, salts, spices. It kind of gets in the way between the yeast and the sugar and the starches that it wants to eat. And so it slows it down. It takes longer. It's harder for the yeast to reach its food source. It can get there — it just takes more time.
>
> So when you're cooking with added fat, salt, spices to a dough, you just need to factor in that it's going to take more time for it to proof, for it to get to what it needs. Because it needs starches, sugars, it needs water. That's all it needs.
>
> A pre-ferment is flour, water, yeast. That's all that's needed.
>
> I know some yeast says it needs sugar to activate — active dry yeast, they recommend that. Really, if you give it time, you don't need the sugar in it. You don't have to add the sugar to your dough, unless you want a sweeter dough.
>
> So it's just: how much time do you want? If you need to bake dough and you want it to rise a little bit faster, you can add sugar to it. It's just going to be a sweeter bread — but it will rise a little faster, get going a little faster, shorten your proofing time.

Picture — the failure-then-fix shot, and the most useful one on the whole list. Two doughs, same flour, same clock, side by side: one lean, one with butter and salt worked in. A locked-off timelapse over three or four hours. It proves in six seconds what she spends two minutes explaining, and it is the shot most likely to be screenshotted.

B-roll. Fat, salt and spice going into a bowl, shot close. And the pre-ferment being mixed — flour, water, yeast, nothing else — because she says "that's all it needs" and the shot can be literally that: three things and no fourth.

Graphic. A small comparison card for the sugar question — same two jars, one risen further, with the trade-off lettered underneath. Keep it to two items; §12 fails more than five.

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9. So why are there so many kinds? — 7:31

> *Saccharomyces cerevisiae* — I said it's all the same thing. But why? Why do we have so many different forms, when it's all the same guy?
>
> Well, he likes to go dormant. He might die. He needs access to food sources. It comes down to how frequently does the person using it need to have access to the yeast.
>
> A baker is using yeast day in, day out. They can have fresh yeast on hand, because they might not want to wait for a dry yeast to come building back up into active yeast that is moving and flowing. Having fresh yeast will shorten their cook time, their baking time, their proof time, and it makes it easier.
>
> But if you have a baker in Kansas that lives on a farm that takes an hour to get to their local grocery store — having fresh yeast is not practical, because they need something that can stay on hand, that lasts longer. That's where you get your active dry and your instant yeast. It's a casual baker, a person that doesn't want something that is gonna go and die within a couple weeks, a week. They need something that lasts longer, because of the effort it takes to get to the grocery store.
>
> Also, those grocery stores probably don't sell a lot of yeast, because they might not have as frequent customers. And so having fresh yeast on hand, and having it die on hand again and again, might not be financially viable. So they have the dry yeast on hand, so that way when the customer comes in it's available, it's not dead. It's what those people need.
>
> But if you have yeast on hand, and you have someone that's wanting to learn how to brew, they're searching — how do I find yeast? And they get brewer's yeast. "Yes, I'm brewing, let me buy brewer's yeast." It's marketing sometimes. Sometimes it's convenience. Sometimes it's who the customer is. But if you know what you're looking for, you can buy what you can find and work with it.

Picture. Her to camera for eight minutes of runtime by this point. This section needs the most help and has the least obvious imagery.

Graphic — the two bakers. A working bakery on one side, a farmhouse an hour from a shop on the other, and the same organism sold in two different forms between them. Drawn as a pair of cards rather than photographed; it is an argument, not a place.

B-roll. A supermarket baking aisle — the actual shelf, with the actual small yeast section — is a cheap shot that lands her point about shelf economics better than any drawing.

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10. Instant, active dry, and glutathione — 9:31

> If you're really wanting to deep dive into making the best, most perfect bread, the most premium bread — you might care what type of yeast, because there is a little bit of a difference with instant and active dry.
>
> And so the yeast might become damaged from sitting and being dried out. And when it comes up and back to life, it can release something called glutathione. And what that can do is, it can get into the gluten matrix and attack it a little bit.

> So you have your instant dry yeast — what you might find is, it's available, it's findable, but one of the offsets might be that the bread is not as stretchy. The bread might dry out a little faster, because the starch has a harder time holding onto water with the GSH involved.
>
> And so day to day, does it really matter? Probably not, with the bread that you want to make. But if it really matters — if you're finding that your bread is drying out, it's not as stretchy, it's not as elastic as you want the bite to be — try seeing if you can find fresh yeast and switching to that. That might help fix the problem that you run into.

⚠ The glutathione mechanism, and the claim that it makes bread dry out faster via starch holding water. ⚠ She introduces the abbreviation **GSH** in a take that isn't in the cut, then uses it here as though it had been introduced — so as cut, "GSH" arrives unexplained. Fix with a label graphic rather than a re-record.

Graphic — glutathione. The molecular kit's second real job, and it needs **GSH** on a label card because she uses the abbreviation in the cut without ever introducing it (the take where she does is unusable).

Graphic — the gluten matrix. Long strands, then strands cut short. This is the clearest before/after in the video and it should read at a glance: the same picture twice, one of them snipped.

Picture — windowpane, both doughs. Stretch each until it tears or doesn't. Then cut both loaves and shoot the crumb. Again: failure, then fix. **Blocked on D6** for the mechanism.

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11. Use what you can get — 10:56

> But if you're happy with your bread, it doesn't matter if you have access to fresh yeast. Active dry works great. If you have fresh yeast available, but you can't find, for some reason, the active dry or instant yeast — go ahead and use that. Use what's available. It's all the same.
>
> It's what we've done for hundreds of years. I mean, look at our ancestors who made sourdough. Sourdough was what was available. They didn't have to ship yeast from back east to the hard to reach people. And so that's why, for some people, sourdough is even better than active dry yeast for them.
>
> And if you're struggling because you want a fresh yeast, and all you can find is active dry — try just making sourdough from your active dry. Get it going, get it started, let the yeast multiply, be happy for a few days, and then you have a little pre-ferment already made, and go from there.

B-roll. Sourdough going west is a map card, and it is the one moment where a period photograph would be forgivable if the licensing is clean. Otherwise: hands tearing a piece off a mother and passing it over, which also closes the loop with §3.

Picture. The active-dry-to-sourdough conversion she describes is a four-day timelapse of one jar. Worth starting *now*, before the reshoot, because it takes real time.

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12. Close — 11:56

> It can be played with. Yeast is not something that has to be a hard and fast rule. It feels like a mystery, but really all it is is the same little *Saccharomyces cerevisiae* bug, again and again and again and again. He's happy.
>
> All the little bug wants is to go and eat sugars and fart — produce carbon dioxide and ethanol. It just wants to live a happy life, and it's not all that complicated.

⚠ "…eat sugars and fart" is in the audio at 12:20 and two separate transcription passes disagreed about it, so it needs an ear rather than a machine. It is the best line in the video and it is also a decision about what this channel sounds like — keep it, or cut straight to "produce carbon dioxide and ethanol", which is a clean edit either way.

⚠ There is no outro *written*. The outro **music** starts at 11:51 and is placed so it lands its own ending on the last frame, which gives this section a lift it doesn't currently earn — §9 asks for one takeaway and one next action, under 20 seconds, and this ends on a thought rather than on an ask. The music is now holding a shape the script hasn't filled.

Picture — the outro music is already here and the picture isn't. The song lands its ending on the last frame, which gives this section a shape nothing on screen currently fills.

The obvious close is the **jar from the very first shot**, still going — the same jar, hours later, which makes the twelve minutes feel like one sitting. Shoot it at the same time as the hook macro so the light matches.

Graphic. One end card. §9 says under 20 seconds, one takeaway, one next action — and there is no next action written yet (C6).

What was left out, and why

MVI_0018 (4:23) — nothing used. It is an earlier, looser pass at the same history that MVI_0019 covers better, and she says on tape "*I'm skipping the introduction part, because a lot of it's just voiceovers that I'll do later.*" Two things in it are worth keeping as alternates if a line needs replacing: the bakers-and-brewers exchange (2:22–2:31) and the sourdough-mother analogy (3:52–4:02), which is warmer than the version in MVI_0021.

MVI_0020 (3.8 s) is one line — "What does that say?" — and is not footage.

Everything with Nathan in it. He is on the camera mic through MVI_0018 and MVI_0019, and several of her best lines are answers to a question he asked. None of the questions are in the cut, so a few paragraphs start slightly cold.

The abandoned takes. She restarts a lot, which is normal and is what the retake rules exist for: the last complete attempt survives, the earlier ones are cut, and every join carries a marker in the FCPXML saying which take it is and why it's there.

Four sequences worth having but not usable as recorded — the enzyme cascade middle, the "that's why we use yeast in food" payoff, the alcohol-stays-in-wine contrast, and the sourdough-went-west

passage. All four are listed as pickups in `build-sheet.md`.